

STATISTICAL RESEARCH: DATA SOURCES

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What one observation represents

Each scoring row represents an NHL **player and regular season**. We join five-on-five NHL events and shifts to the same players and games in the [All Three Zones transition workbook](#). Behavior spans 2021–22 through 2024–25; roster outcomes extend through 2025–26 for a one-year follow-up.

NHL play-by-play supplies contact and shooting events, rosters identify players, and shifts measure five-on-five exposure. Corey Sznajder’s manually tracked records add forechecking, retrieval, and entry-defense actions. We retrieve NHL records through the pinned [nhlscraper](#) revision and follow the [All Three Zones glossary](#). The workbook was retrieved on 2026-09-15; its SHA-256 fingerprint is 0ed9c09819854e6f4b281730ba2d0b3e2dcc629a48ce6fc6fbf9bc35331907fd.

Official NHL draft-year scouting passages provide an earlier description of physical traits. Public files retain coded mentions and a [source catalog](#); passage text and blinded identity keys stay outside Git.

Matching and eligibility

We preserve the workbook’s original labels while matching games by teams, a date within three days, and NHL rosters. Names and sweater numbers help resolve players who share a name. A game match must be unambiguous and identify at least 90% of its players. We retain one valid row per NHL game and player, collapsing identical duplicates and excluding conflicts.

The workbook contains 1,699 game labels. Of these, 1,684 resolve to 1,683 distinct NHL games. Fifteen labels remain unresolved. At the player-game level, the matched audit yields the following flow.

Table 1: Disposition of All Three Zones skater-game rows after identity, count, and exposure checks.

Status	Rows
Retained	60,331
Unresolved game	540
Unresolved player	1
Invalid event counts	6
Player exposure disagreement	40
Game exposure disagreement	180

We compare source ice time with NHL five-on-five exposure reconstructed from shift intervals. A player-game is set aside when its difference exceeds 120 seconds, or its game’s median absolute difference exceeds 60 seconds. Counts must be nonnegative integers, and denials cannot exceed targets. These checks leave 60,331 player-game rows; their median absolute exposure difference is 11.00 seconds. Rates use **NHL shift-derived minutes**, while source minutes remain for audit.

Player-seasons need at least 300 full-season NHL minutes, listed height and weight, and a known birth date. Scoring also requires 150 matched five-on-five minutes. Of 2,731 candidate player-seasons, 2,314 meet that threshold: 1,485 forwards and 829 defensemen. The other 417 lack enough tracked exposure. Published rankings will require 500 full-season minutes.

Table 2: Eligible scored player-seasons by position and behavior season.

Season	Forwards	Defensemen	Total
2021–22	399	215	614
2022–23	374	203	577
2023–24	359	204	563
2024–25	353	207	560

The rows represent 825 players. Median matched exposure is 280.92 minutes for forwards and 351.18 for defensemen. Each season is modeled separately.

Variables used to construct CSAx

The six direct measures appear in both positional models. Each counts a player’s attributed NHL event in a matched five-on-five game and divides by that player’s NHL five-on-five minutes, expressed per 60. A recorded zero event is kept as zero when exposure is positive.

Measure	Recorded numerator and interpretation
Hits delivered	Hits attributed to the player as hitter; possession and role affect opportunities.
Hits received	Hits attributed to the player as recipient; exposure to contact need not mean initiating it.
Opponent shots blocked	Shots blocked for the defending team, excluding teammate blocks; a recorded block need not involve body contact.
Fights	Recorded fighting infractions, counted separately from contact penalties; events are sparse.
Contact penalties taken	Whitelisted contact-related infractions attributed to the player, counted as incidents rather than minutes.
Contact penalties drawn	Whitelisted infractions committed by opponents against the player; puck possession and skill also affect exposure.

The penalty whitelist covers boarding, charging, checking from behind, clipping, elbowing, head checks, kneeing, roughing, slew-footing, cross-checking, high-sticking, holding, holding the stick, hooking, and tripping, with recorded helmet-removal and double-minor variants. Fighting stays separate. Exact source labels are fixed in `R/functions.R`.

Forwards and defensemen have different indirect opportunities. These variables describe involvement around contested space; none directly records pressure on the player at every moment.

Forward measure	Unit and denominator	Interpretation
Net-front attempt share	Located unblocked attempts at $82 \leq x \leq 89$ and $ y \leq 8$ feet / all located unblocked attempts	Shooting near the crease; location does not confirm pressure.
Tip/deflection share	Tips and deflections / shots on goal or goals with known shot type	Redirections often occur near traffic; role affects opportunities.
Median shot distance	Feet from goal among located unblocked attempts	Proximity of shooting involvement.
Backhand share	Backhands / shots on goal or goals with known shot type	Possible constrained shooting space; backhands occur elsewhere.
Dump-in recoveries	All Three Zones Recoveries per 60 matched minutes	Recovering a dump-in and making a play.
Forecheck pressures	All Three Zones Forecheck Pressures per 60 matched minutes	Pressuring a puck-carrying defender.

Defenseman measure	Unit and denominator	Interpretation
Defensive-zone retrievals	All Three Zones DZ Retrievals per 60 matched minutes	Collecting the puck in the defensive zone and making a play.
Botched retrievals	All Three Zones Botched Retrievals per 60 matched minutes	Retrieval-related breakdowns; the tracked category can involve the receiving player.
Entry denial share	All Three Zones Denials divided by Targets	Preventing an opponent's attempted entry; gap control and team tactics also matter.

Source counts and share denominators remain in the feature table. A share with zero opportunities stays missing before training-sample imputation. One center-season has an undefined entry-denial share; the forward model does not use that feature. All modeled primary features are present. Twenty opportunities is a caution threshold for sparse denominators, not a reliability guarantee.

Outcomes and interpretation

The scouting code records mentions of active physical traits such as checking and puck battles in an earlier report. No mention is not evidence of soft play. Official NHL records provide power-play and penalty-kill minutes as shares of total regular-season ice time; recorded zeros remain distinct from missing values.

Postseason contact uses five-on-five hits, hits received, and opponent shots blocked in each team's first four playoff games, compared with regular-season games excluded from CSAx inputs. Rate comparisons require positive exposure in both periods. Continuation means at least 300 NHL minutes next season; minutes per game is defined among returners. Adjustment variables are listed size, age, current games and ice time per game, five-on-five points per 60, relative shot attempts, and season.

Tracked games are unevenly selected. Player-game totals have no timestamps or linked possession sequences, so retrievals and denials remain proxies for physical involvement. Interpretation depends on scouting evidence, role, and tracking coverage.

Reproduction and sources

Bundled `data/analysis_data.rds` supplies inputs and provenance. Restore `renv.lock` and render this QMD with Quarto; fresh acquisition requires the [All Three Zones workbook](#) and pinned `nhlscraper`.